

Indoor balance-challenge cards

Four indoor challenges that replace a real ropes course. Do EVERY challenge next to a wall so a slip just leaves you leaning. Keep the floor clear; a helper spots the chair stand; the taped line stays flat (no running, no raised beam). Predict first, then test, then map the forces.

1 Wall and Heel

Stand with your back and heels flat against a wall. Try to bend and touch your toes, or lift one foot forward, without leaning off the wall.

Predict: will it WORK or FAIL? Where does your center of mass go? _____

Center-of-mass reason: It fails: the wall stops you leaning forward, so your center of mass cannot move over your toes (your base).

2 Backless Chair Stand

Sit tall on a backless chair, feet flat, arms crossed. Try to stand up WITHOUT leaning your chest forward.

Predict: will it WORK or FAIL? Where does your center of mass go? _____

Center-of-mass reason: It fails: you must shift your center of mass forward over your feet first. No lean = no balance over the base.

3 Toe Grab Hop

Bend down, grab your toes, and try to hop forward while still holding them.

Predict: will it WORK or FAIL? Where does your center of mass go? _____

Center-of-mass reason: It is very hard: holding your toes locks your center of mass low and forward, so you cannot get your base under you to jump.

4 Loaded Line Walk

Walk a taped floor line holding a weight straight out at arm's length, then again holding it at your chest. Which is steadier?

Predict: will it WORK or FAIL? Where does your center of mass go? _____

Center-of-mass reason: Arm's length moves your center of mass sideways and forward off the line; held at the chest it stays over your base, so you wobble less.